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Efficient Factor-Based Conditional Diffusion Model for Contextual Portfolio Optimization

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1 Introduction

Diffusion-based generative models are a powerful class of probabilistic generative AI models capable of generating samples from high-dimensional target data distributions given a set of training data [Song and Ermon, 2019, Ho et al., 2020]. The core idea is to use a forward process to gradually corrupt the unknown target distribution into a simple noise distribution, then learn to reverse this process to generate new samples. Diffusion models have achieved state-of-the-art performance across a wide range of domains, including image and audio generation [Dhariwal and Nichol, 2021, Rombach et al., 2022, Ramesh et al., 2022], and have recently been extended to time series forecasting [Rasul et al., 2021, Li et al., 2022]. Despite these advances, the application of diffusion models to financial decision-making problems such as portfolio optimization remains relatively underexplored.

Gao et al. [2025] propose a factor-based conditional diffusion model for contextual portfolio optimization that generates the full cross-sectional distribution of next-day stock returns conditioned on asset-specific observable factors. Their model adapts the Diffusion Transformer (DiT) architecture [Peebles and Xie, 2023] with three key modifications: operating directly in raw return space without a variational autoencoder, employing token-wise conditioning so that each asset’s return is linked to its own factor vector, and computing per-token AdaLN-Zero modulation parameters independently for each asset. Empirical results on the Chinese A-share market demonstrate that mean-variance portfolios constructed from their generated return distributions consistently outperform those based on empirical, shrinkage, and DCC-GARCH estimators. However, a critical practical limitation of this approach is its computational cost at inference time: the standard DDPM reverse process requires on the order of $N = 1000$ denoising steps per sample, and generating the $M = 500$ – 1000 return samples needed for stable moment estimation on every trading day is computationally prohibitive, even on modern hardware.

In this work, we replicate and extend Gao et al. [2025] with a focus on inference efficiency, adapting the methodology to the US equity market using S&P 500 constituents as a natural analog to the original CSI 300 universe. Our central contribution is the introduction of Denoising Diffusion Implicit Models (DDIM; Song et al. 2021) as a drop-in replacement for the DDPM reverse process at inference time. DDIM decouples the number of sampling steps from the number of training steps, enabling high-quality sample generation in as few as 50–100 steps

without any retraining of the underlying model. This yields substantial wall-clock speedups at daily rebalancing frequency while preserving the portfolio performance characteristics of the original approach. Additionally, we contribute an open-source, reproducible data pipeline for price-volume factor construction on the US equity market using publically available OHLCV data and the WorldQuant Alpha 101 factor library [Kakushadze, 2016], providing a standardized benchmark dataset for future research in factor-based generative modeling.

Contributions. Our contributions can be summarized in the following points.

1. We introduce DDIM sampling as a competitive method to generating realistic paths for portfolio optimization.
2. We introduce a canonical open source dataset for technical factor based research.

2 Methodology

2.1 Problem Formulation

Following the Arbitrage Pricing Theory [Ross, 2013] and the general asset pricing framework of Gu et al. [2020], we model stock returns using the statistical factor model:

$$R_{t+1} = f(X_t) + u_{t+1}, \quad (1)$$

where $R_{t+1} \in \mathbb{R}^D$ is the return vector of D stocks from time t to $t+1$, $X_t = (x_{1,t}, x_{2,t}, \dots, x_{D,t})^\top \in \mathbb{R}^{D \times K}$ is a factor matrix observable at time t , u_{t+1} is a random shock independent of information at time t , and f is an unknown deterministic function describing the dependence of returns on factors. Since both f and the distribution of u_{t+1} are unknown, the conditional distribution of R_{t+1} given X_t must be learned from data.

In practice, we have access to a set of observations $\{(x_t, r_{t+1})\}_{t=0}^{T-1}$ as T samples of (X_t, R_{t+1}) . A key challenge is that for any specific factor realization x , there are no historical return observations that directly correspond to $P_{R_{t+1}|X_t=x}$, since the factor matrix changes every day. Classical non-parametric density estimators are therefore inapplicable, particularly when both the return space ($D \approx 500$) and the factor space ($K = 101$) are high-dimensional. We overcome this by leveraging a conditional diffusion model to learn $P_{R_{t+1}|X_t=x}$ for an arbitrary factor matrix x , which can then generate an arbitrary number of synthetic return samples from the

desired conditional distribution. These generated samples $\{\hat{r}_{t+1}^{(j)}\}_{j=1}^M$ are used to estimate the conditional mean and covariance as sample moments:

$$\hat{\mu}_{t+1} = \frac{1}{M} \sum_{j=1}^M \hat{r}_{t+1}^{(j)}, \quad \hat{\Sigma}_{t+1} = \frac{1}{M-1} \sum_{j=1}^M \left(\hat{r}_{t+1}^{(j)} - \hat{\mu}_{t+1} \right) \left(\hat{r}_{t+1}^{(j)} - \hat{\mu}_{t+1} \right)^\top. \quad (2)$$

In practice, finite-sample estimation under bfloat16 inference can introduce small numerical asymmetries and near-singularity in $\hat{\Sigma}_{t+1}$, especially when the number of samples is similar to the number of assets. We, therefore, implement the robust covariance estimator developed in Ledoit and Wolf [2004].

Given the estimated moments $(\hat{\mu}_{t+1}, \hat{\Sigma}_{t+1})$, the investor solves the contextual mean-variance portfolio optimization problem:

$$\max_{\omega \in \mathcal{W}} \hat{\mu}_{t+1}^\top \omega - \frac{\gamma}{2} \omega^\top \hat{\Sigma}_{t+1} \omega, \quad (3)$$

where $\omega = (\omega_1, \dots, \omega_D)^\top$ is the portfolio weight vector, $\mathcal{W} = \{\omega \in \mathbb{R}^D : \mathbf{1}^\top \omega = 1, \omega_i \geq 0\}$ enforces a fully-invested long-only constraint reflecting realistic institutional constraints in US equity markets, and $\gamma > 0$ is the risk aversion parameter. To account for transaction costs incurred by daily rebalancing, we follow Gao et al. [2025] and consider the augmented problem:

$$\max_{\omega, b, s} \omega^\top \hat{\mu}_{t+1} - \frac{\gamma}{2} \omega^\top \hat{\Sigma}_{t+1} \omega - 0.001 (b^\top \mathbf{1} + s^\top \mathbf{1}) \quad (4)$$

$$\text{s.t. } \omega^\top \mathbf{1} = 1, \quad 0 \leq \omega_i \leq 1, \quad b \geq 0, \quad s \geq 0, \quad \omega_i - \omega_i^d = b_i - s_i, \quad \forall i,$$

where b and s are the vectors of buying and selling amounts per unit wealth due to rebalancing, and ω^d denotes the portfolio weight vector immediately before rebalancing. We assume somewhat realistic transaction costs of 10 bps per unit transaction, loosely based off Frazzini et al. [2018]. This quadratic program is solved daily using `cvxpy` [Diamond and Boyd, 2016].

2.2 Diffusion Models

We adopt the conditional Denoising Diffusion Probabilistic Model (DDPM; Ho et al. 2020) to learn the distribution of R_{t+1} given X_t . Given a training pair (X_t, R_{t+1}) , the forward diffusion

process gradually corrupts the return vector by adding Gaussian noise over N steps:

$$R_{t+1}^{(0)} = R_{t+1}, \quad R_{t+1}^{(n)} = \sqrt{1 - \beta_n} R_{t+1}^{(n-1)} + \sqrt{\beta_n} \epsilon_n, \quad n = 1, \dots, N, \quad (5)$$

where $\epsilon_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_D)$ and $\{\beta_n\}_{n=1}^N \in (0, 1)$ is a predefined variance schedule. Defining $\alpha_n = 1 - \beta_n$ and $\bar{\alpha}_n = \prod_{s=1}^n \alpha_s$, the forward process admits the closed-form marginal:

$$R_{t+1}^{(n)} = \sqrt{\bar{\alpha}_n} R_{t+1}^{(0)} + \sqrt{1 - \bar{\alpha}_n} \bar{\epsilon}, \quad \bar{\epsilon} \sim \mathcal{N}(0, I_D), \quad (6)$$

which allows sampling $R_{t+1}^{(n)}$ at any step n directly from the original return $R_{t+1}^{(0)}$ without iterating through the chain.

In the reverse (denoising) process, we initialize $R_{t+1}^{(N)} \sim \mathcal{N}(0, I_D)$ and iteratively sample:

$$R_{t+1}^{(n-1)} \sim \mathcal{N}\left(\frac{1}{\sqrt{\alpha_n}} \left(R_{t+1}^{(n)} - \frac{\beta_n}{\sqrt{1 - \bar{\alpha}_n}} \epsilon_\theta(R_{t+1}^{(n)}, n; X_t)\right), \sigma_n^2 I_D\right), \quad n = N, \dots, 1, \quad (7)$$

where $\sigma_n^2 = \frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n} \beta_n$ and $\epsilon_\theta(\cdot)$ is a neural network with parameters θ trained to predict the noise added in the forward process. The network is trained by minimizing the noise prediction loss:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \mathbb{E} \left[\left\| \epsilon - \epsilon_\theta(R_{t+1}^{(n)}, n; X_t) \right\|^2 \right], \quad (8)$$

where the expectation is taken over (X_t, R_{t+1}) drawn from the training data and $\epsilon \sim \mathcal{N}(0, I_D)$. Once trained, the reverse process is run to generate synthetic return samples from the learned conditional distribution $P_{R_{t+1}|X_t=x}$, from which the mean μ_{t+1} and covariance Σ_{t+1} are estimated as sample moments and fed directly into the portfolio optimization problem.

2.3 DDIM Sampling

A compelling alternative to DDPM sampling is DDIM sampling Song et al. [2021]. DDIM sampling allows significantly fewer sampling steps to be taken, and as such massively increases the efficiency of generating samples. Additionally, DDIM sampling does not require retraining the DDPM model, but is rather just an alternative, but equivalent method of sampling from the same model.

DDIM defines a non-Markovian forward process that shares the same marginals as DDPM

but admits a generalized reverse update:

$$R_{t+1}^{(n-1)} = \sqrt{\bar{\alpha}_{n-1}} \left(\frac{R_{t+1}^{(n)} - \sqrt{1 - \bar{\alpha}_n} \epsilon_\theta(R_{t+1}^{(n)}, n; X_t)}{\sqrt{\bar{\alpha}_n}} \right) + \sqrt{1 - \bar{\alpha}_{n-1} - \sigma_n^2} \epsilon_\theta(R_{t+1}^{(n)}, n; X_t) + \sigma_n \epsilon, \quad (9)$$

where $\epsilon \sim \mathcal{N}(0, I_D)$ and $\sigma_n = \eta \sqrt{(1 - \bar{\alpha}_{n-1}) / (1 - \bar{\alpha}_n)} \sqrt{1 - \bar{\alpha}_n / \bar{\alpha}_{n-1}}$ for a user-specified parameter $\eta \in [0, 1]$. Crucially, DDIM traverses a subsequence $\{\tau_1, \dots, \tau_S\} \subset \{1, \dots, N\}$ of length $S \ll N$, reducing the number of neural network evaluations per sample to S with minimal degradation in sample quality. In our implementation we use the DDIM sampler from the OpenAI DiT codebase [Peebles and Xie, 2023], and support both DDPM and DDIM sampling via a configurable `sample_method` argument.

2.4 Architecture

Our denoising network ϵ_θ adapts the Diffusion Transformer (DiT; Peebles and Xie 2023) architecture to the financial cross-sectional return setting, following the three key modifications introduced by Gao et al. [2025] and implemented using the facebookresearch DiT codebase as a foundation.

Tokenization. Each asset i ’s scalar noisy return $R_{t+1,i}^{(n)}$ is mapped to a d_{model} -dimensional embedding via a learned linear projection with bias (the `ReturnsEmbedder`), producing a sequence of D tokens $\{h_i\}_{i=1}^D$. This eliminates the need for a Variational Autoencoder (VAE) used in the original DiT for image patches, since stock returns are already low-dimensional.

Per-token conditioning. The diffusion timestep n is encoded into a vector $e_n \in \mathbb{R}^{d_{\text{model}}}$ via a sinusoidal timestep embedder. Simultaneously, each asset’s factor vector $x_{i,t} \in \mathbb{R}^K$ is processed by a two-layer MLP with SiLU activation (the `FactorMLP`) to produce a factor embedding of the same dimension. These are summed element-wise to form the per-token condition vector:

$$c_i = \text{MLP}_{\text{factor}}(x_{i,t}) + e_n \in \mathbb{R}^{d_{\text{model}}}, \quad (10)$$

so that each asset carries its own conditioning signal encoding both its factor information and the current diffusion step.

DiT blocks. The sequence of return embeddings is passed through L identical DiT blocks. In each block, per-token AdaLN-Zero modulation parameters $(\gamma_{i,1}, \beta_{i,1}, \alpha_{i,1}, \gamma_{i,2}, \beta_{i,2}, \alpha_{i,2})$ are

computed independently for each token from its condition vector c_i via a shared MLP, initialized to zero following Peebles and Xie [2023]. The block updates proceed as:

$$\hat{h}_i^\ell = h_i^\ell + \alpha_{i,1} \cdot \text{MHSA}\left((1 + \gamma_{i,1}) \odot \text{LN}(h_i^\ell) + \beta_{i,1}\right), \quad (11)$$

$$h_i^{\ell+1} = \hat{h}_i^\ell + \alpha_{i,2} \cdot \text{FFN}\left((1 + \gamma_{i,2}) \odot \text{LN}(\hat{h}_i^\ell) + \beta_{i,2}\right), \quad (12)$$

where $\text{LN}(\cdot)$ denotes Layer Normalization [Ba et al., 2016] and $\odot$ is the element-wise product. The Multi-Head Self-Attention (MHSA) layer allows every asset’s embedding to interact with every other asset’s embedding, capturing cross-asset return dependencies globally, while the per-token modulation accounts for asset-level heterogeneity introduced by differing factor vectors.

Output. After the final DiT block, a **FinalLayer** applies a further AdaLN-Zero normalization followed by a linear projection from d_{model} back to a scalar, producing the predicted noise vector $\epsilon_\theta(R_{t+1}^{(n)}, n; X_t) \in \mathbb{R}^D$.

The architecture is summarized in 1, which is adapted from Gao et al. [2025].

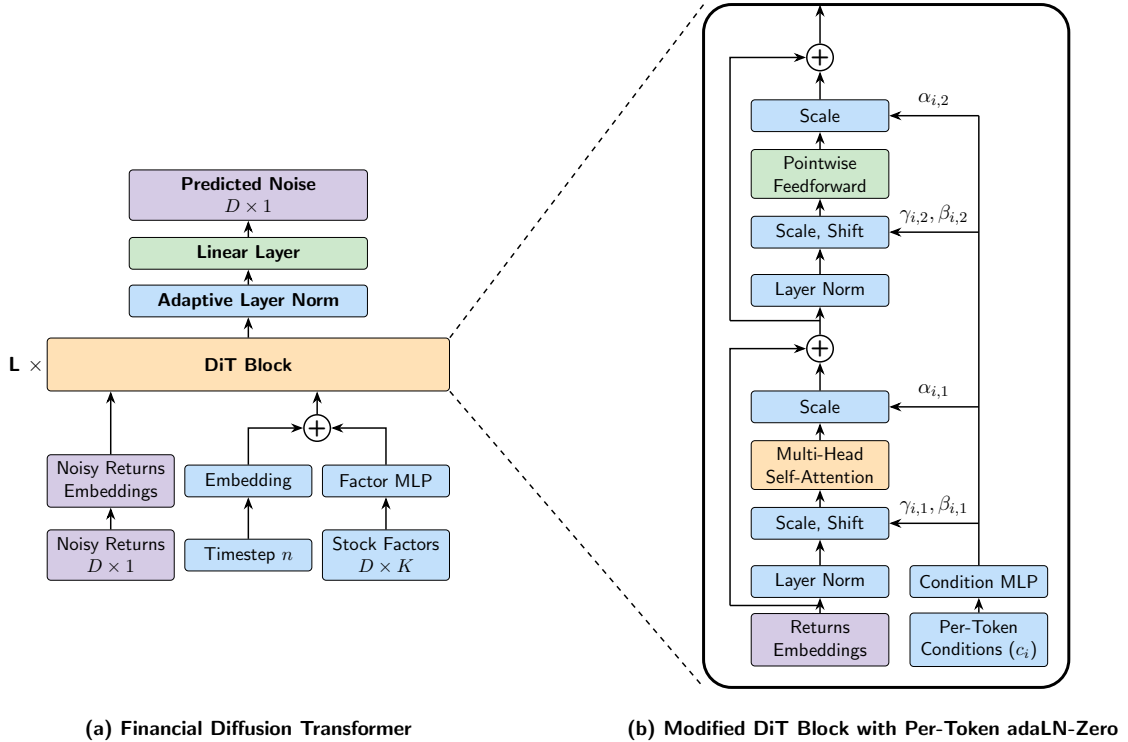


Figure 1: Factor DiT

We provide four model configurations of varying capacity, summarized in Table 1, ranging from FactorDiT-S to FactorDiT-XL, following the scaling convention of Peebles and Xie [2023].

Table 1: FactorDiT model configurations.

Model	Layers L	Hidden size d_{model}	Attention heads
FactorDiT-S	12	384	6
FactorDiT-B	12	768	12
FactorDiT-L	24	1024	16
FactorDiT-XL	28	1152	16

3 Experimental Results

3.1 Data and Experimental Setup

Asset universe. We consider daily return data for S&P 500 constituents as the asset universe, sourced from the CRSP daily stock file via WRDS. To avoid look-ahead bias, we fix the universe to stocks that appear in the S&P 500 constituent list and maintain complete daily observations throughout the full sample period, yielding $D \approx 500$ stocks. Raw returns are winsorized cross-sectionally at three standard deviations on each trading day to mitigate the influence of outliers.

Factor construction. As the 208 proprietary price-volume factors used in the original work were inaccessible to us, we instead employ the 101 formulaic alphas of Kakushadze [2016] as our conditioning factor set ($K = 101$). These alphas are constructed entirely from publicly available price and volume data (open, high, low, close, volume, and turnover), making our pipeline fully open-source and reproducible. Factors are computed daily in a point-in-time manner using only information available up to and including day t , standardized cross-sectionally to zero mean and unit variance, and winsorized at three standard deviations to match the preprocessing of Gao et al. [2025].

Dataset split. The dataset is split chronologically into a training period from January 2017 to May 2023 and a test period from May 2023 to December 2024, maintaining an approximately 80:20 train/test ratio. Pre-2017 data is used as a warmup period for rolling-window factor computations and is discarded before training. To avoid look-ahead bias, all preprocessing statistics (e.g., columns identified as degenerate) are computed using training data only.

Training. We follow the basic training script given in the DiT implementation. The small model is trained for 1400 epochs using the AdamW optimizer with a learning rate of 10^{-4} and a batch size of 256, on an NVIDIA A100-80GB GPU. We train with a cosine noise schedule

over $N = 1000$ diffusion steps. An exponential moving average (EMA) of the model weights is maintained throughout training and used exclusively at inference time.

Inference and portfolio construction. During the test period, on each trading day t we generate M synthetic return samples $\{\hat{r}_{t+1}^{(j)}\}_{j=1}^M$ from the trained model conditioned on the current factor matrix X_t , using either DDPM or DDIM sampling. The estimated moments $(\hat{\mu}_{t+1}, \hat{\Sigma}_{t+1})$ are then used to solve the mean-variance optimization problem described in Section 2.1, and the resulting optimal weights are implemented for the next trading day.

3.2 Portfolio optimization

We compare our DDIM approach against our DDPM implementation of Gao et al. [2025] for this market. Our main goal is to make sure that for every performance metric, Sharpe Ratio, Sortino Ratio, Calmar Ratio, and Return-to-CVar, the DDIM sampler *does not perform worse, while achieving better sample efficiency*. As we can see in Figure 2, we indeed achieve this goal. We attain near identical performance, while sampling 70% faster. This is quite crucial, as for the larger sample sizes, we required nearly 7 hours to generate 2500 samples with the DDPM. As the eventual goal is to generate samples for the entire market, more than 5000 assets, this high number of samples is crucial, or we have very poor estimates for our covariance matrix. It’s clear that with DDIM sampling, we can overcome this intractability, and eventually develop the method for the entire market.

		Seconds	Sharpe	Sortino	Calmar	RtC
Sampler	Samples					
DDIM	100	300.96	1.13	1.16	1.33	0.67
	500	1510.47	1.20	1.23	1.43	0.71
	1000	3022.64	1.21	1.23	1.43	0.71
	2500	7556.66	1.20	1.23	1.43	0.71
DDPM	100	977.81	1.13	1.16	1.32	0.67
	500	4896.27	1.20	1.22	1.42	0.71
	1000	9804.65	1.20	1.23	1.42	0.71
	2500	24508.24	1.19	1.22	1.41	0.71

Figure 2: Portfolio Optimization Results

4 Conclusion

There are several directions to future research. A first step is expanding this method to the entire market, as opposed to only S&P 500 constituents. Additionally, adapting the Stable Diffusion 3 MMDiT presented in Esser et al. [2024] architecture and flow matching loss would be a very natural step in applying modern diffusion techniques to finance. Flow models have even faster inference times than DDIMs, and produce higher fidelity samples. Additionally, we trained our model without Classifier-Free-Guidance (CFG). This is a technique presented in Ho and Salimans [2022] used in essentially all modern generative image models to increase the fidelity of class conditional generated samples. Implementing CFG could boost the fidelity of our generated samples and boost our portfolio’s performance. Our model focuses on one-step forecasting. One step forecasting is a hugely important task, but multi-step forecasting is just as important. A method to allow efficient training of a multi-step forecasting model is a very important avenue of further research. Additionally, the only factors currently included are technical factors. Fundamental factors, market regimes, news, or other sources of sentiment could be included in future work. This may be non-trivial, since this data is available in non-regular intervals. Finally, moving away from hand-made factors would be an ideal end goal, perhaps utilizing a pre-trained VAE for financial data.

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